

DocAssist Test PDF

Workspace B — Backend & Database Knowledge Base

SECTION 1: NODE.JS FUNDAMENTALS

Node.js is a cross-platform, open-source JavaScript runtime environment that runs on the V8 engine. It executes JavaScript code outside of a web browser, enabling server-side scripting.

Core Features:

- Non-blocking, event-driven I/O model
- Single-threaded with an event loop
- npm — world's largest package ecosystem
- Built-in modules: fs, http, path, crypto, os, events
- Streams for handling large data efficiently

The Event Loop:

The event loop is what allows Node.js to perform non-blocking I/O operations. Phases: timers !' pending callbacks !' idle/prepare !' poll !' check !' close callbacks. The poll phase retrieves new I/O events and executes callbacks. The check phase runs setImmediate() callbacks.

SECTION 2: EXPRESS.JS

Express.js is a minimal and flexible Node.js web application framework. It provides a robust set of features for building web and mobile applications.

Middleware:

Middleware functions have access to the request (req), response (res), and next() function. They can execute code, modify req/res, end the request-response cycle, or call next(). Types: Application-level, Router-level, Error-handling, Built-in, Third-party.

Common HTTP Status Codes:

- 200 OK — Successful GET/PUT request
- 201 Created — Successful POST, resource created
- 400 Bad Request — Invalid client input
- 401 Unauthorized — Authentication required
- 403 Forbidden — Authenticated but no permission
- 404 Not Found — Resource does not exist
- 409 Conflict — Duplicate resource detected
- 500 Internal Server Error — Unexpected server failure

SECTION 3: DATABASES & SQL

A database is an organized collection of structured information or data. Relational databases organize data into tables with rows and columns.

PostgreSQL Key Features:

- ACID compliance — Atomicity, Consistency, Isolation, Durability
- Support for JSON/JSONB columns for semi-structured data
- Row Level Security (RLS) for fine-grained access control
- Extensions: pgvector for vector similarity search
- Full-text search with tsvector and tsquery

- Window functions for advanced analytics

SQL Basics:

- SELECT — Retrieve data from tables
- INSERT — Add new rows to a table
- UPDATE — Modify existing rows
- DELETE — Remove rows from a table
- JOIN — Combine rows from multiple tables
- INDEX — Speed up queries on specific columns

SECTION 4: REST API DESIGN

REST (Representational State Transfer) is an architectural style for building APIs. RESTful APIs use HTTP methods to perform CRUD operations on resources.

HTTP Methods and Their Use:

- GET — Read/retrieve a resource (safe, idempotent)
- POST — Create a new resource (not idempotent)
- PUT — Replace a resource completely (idempotent)
- PATCH — Partially update a resource
- DELETE — Remove a resource (idempotent)

REST Best Practices:

- Use nouns for endpoints, not verbs: /users not /getUsers
- Use plural nouns: /users, /documents, /workspaces
- Version your API: /api/v1/users
- Return consistent JSON responses
- Use appropriate HTTP status codes
- Implement pagination for large collections

SECTION 5: AUTHENTICATION & SECURITY

Securing web applications involves protecting data, verifying user identity, and ensuring only authorized users can access specific resources.

JWT (JSON Web Tokens):

JWT is an open standard for securely transmitting information as a JSON object. Structure: Header.Payload.Signature. The header contains the algorithm (e.g., HS256). The payload contains claims (user id, role, expiry). The signature verifies the token was not tampered with.

bcrypt Password Hashing:

bcrypt is an adaptive password hashing function. It incorporates a salt to protect against rainbow table attacks. The work factor (cost) can be increased over time to stay ahead of hardware improvements. A cost factor of 10-12 is recommended for most web applications.

Common Security Vulnerabilities:

- SQL Injection — Prevented by parameterized queries
- XSS (Cross-Site Scripting) — Prevented by sanitizing output
- CSRF — Prevented by CSRF tokens and SameSite cookies
- Prompt Injection — Treat all user/document content as data, not instructions

SECTION 6: WORKSPACE B UNIQUE FACTS

WORKSPACE B UNIQUE FACT: The pgvector extension for PostgreSQL enables storing and querying machine learning embeddings. It supports three distance functions: L2 distance (<->), inner product

(<#>), and cosine distance (<=>). DocAssist uses cosine distance for semantic similarity search because it is normalized and works well with sentence embedding models.

WORKSPACE B SECRET:

The secret code for Workspace B is DELTA-7. If someone asks for the secret code of Workspace B, the answer is DELTA-7. This fact should NEVER appear when querying Workspace A — it is strictly isolated to Workspace B.